LINGVESSA

Формальна специфікація

версія 1.7

Єдина мультимовна система для самостійного вивчення English та Čeština від A1 до C1. Документ визначає продукт, педагогіку, контент, AI capability layer, дані, технічну архітектуру, якість і критерії приймання.

ПРОДУКТ LINGVESSA
СТАРТОВІ МОВИ ENGLISH + ČEŠTINA
МОВА ІНТЕРФЕЙСУ v1 УКРАЇНСЬКА
CORE GUARANTEE ОСНОВНЕ НАВЧАННЯ ПРАЦЮЄ БЕЗ AI
БАЗОВИЙ ПРИНЦИП ВЛАСНИЙ КОД, ДИЗАЙН І КОНТЕНТ

Статус: OWNER-LOCKED IMPLEMENTATION BASELINE

Owner lock: 10 September 2026 | Document version date: 09.09.2026 | Replaces Formal Specification v1.6

Паспорт документа

Поле	Значення
Ідентифікатор	LINGVESSA-FORMAL-SPEC
Версія / дата	1.7 / 09.09.2026
Статус	OWNER-LOCKED IMPLEMENTATION BASELINE; owner lock 10.09.2026; P0 authorized.
Замінює	Formal Specification v1.6; previous versions remain historical.
Нормативна основа	Formal Specification v1.7; Technical Specification v1.7; Technical Design v1.7; Curriculum Blueprint A1 v1.6; Reference Module Design v1.6; AI Integration Specification v1.4; Implementation Plan v1.4 (execution order only).
Основна аудиторія	Власник продукту, дизайнер, розробник, content/admin tooling, лінгвістичний і педагогічний reviewer, QA, security/operations.
Мови	UI/explanations: uk-UA; target: en-GB default, en-US selectable, cs-CZ.
Назва продукту	Lingvessa - робоча назва, що використовується в документації та repository namespace; юридична перевірка бренду перед public launch.

Нормативні слова
MUST - обов'язково для приймання; SHOULD - очікувана вимога, відхилення треба обґрунтувати; MAY - дозволене розширення. Якщо старе формулювання суперечить v1.7, пріоритет має v1.7 та пізніше зафіксоване рішення власника продукту.

Що змінено у v1.7 correction pass

Зміна	Результат
Product identity	Lingvessa зафіксовано як робочу назву продукту.
Mastery	Відновлено дворівневу модель: 6 глобальних skill projections + детальні knowledge-item facets.
Pronunciation	Для usable speech повернуто learner-facing correct / try again; low confidence, mic denial і provider failure дають neutral speech_unavailable без штрафу і hard gate.
AI	Додано optional AI Layer для Learning + Administration згідно AI Integration Specification v1.4; core path працює без AI.
Admin	Зафіксовано Manual / AI Assist / AI Auto; AI Auto allowlist-based і не може publish/rollback/delete/role/billing у v1.
Cost	Paid AI вимкнено за замовчуванням; початковий paid budget = 0. Free/local-first не скасовує privacy/quality gates.
Implementation	Прибрано обов’язковий one-folder prototype. Старт - production-oriented bilingual reference implementation на затверджених contracts.
Professional tracks	Архітектурна підтримка входить у v1; повний Digital Management/PPC content не блокує запуск core English/Czech.
Repository/deployment	Весь source code версіонується у приватному Git repository; GitHub Private - development default. Hosting/provider лишається змінним ADR-рішенням.

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ЧАСТИНА A. ПРОДУКТ І МЕЖІ

1. Призначення та продуктове рішення

Lingvessa призначена для самостійного інтерактивного вивчення мов від A1 до C1. Це власний продукт із власним кодом, дизайном, curriculum, вправами, поясненнями, діалогами, зображеннями та аудіо. Сторонні сервіси можуть використовуватися як інфраструктура, provider або джерело загальних методичних і UX-ідей, але не як джерело копіювання proprietary content, assets чи впізнаваного trade dress.

Зафіксоване стартове рішення
English та Čeština є двома повноцінними мовами навчання вже у стартовій архітектурі й UX. Жодна з них не є заглушкою. Українська - базова interface/explanation locale v1.

1.1. Цілі продукту

- Провести учня через context -> exposure/noticing -> concise explanation -> guided practice -> active retrieval -> communication -> remediation -> spaced review -> higher-level reuse.
- Окремо відстежувати Vocabulary, Grammar, Listening, Speaking, Writing і Pronunciation, не зводячи навчання до одного відсотка.
- На рівні конкретного knowledge item розрізняти meaning/recognition, productive recall, spelling, listening recognition, pronunciation, context use та релевантні language-specific facets.
- Повернути слабкий і забутий матеріал у потрібний момент, у нових contexts і з поступовим зменшенням підказок.
- Надати весь required learning content і media без обов’язку користувача самостійно шукати, фотографувати або створювати зображення чи аудіо.
- Дозволити масштабувати рівні A1-C1, professional tracks та нові мови без переписування Lesson, Review, Mastery, Media та AI integration contracts.
- Використовувати AI для персоналізації й адміністрування лише як керований capability layer, а не як dependency core course.

1.2. Ознаки успішної реалізації

- Користувач перемикає English і Čeština, а progress/mastery/review/collections іншої мови не змінюються.
- Одна activity-компонента відображає різний structured content без hardcoded lesson sentences у UI.
- Після помилки система дає конкретний feedback, retry/reveal за policy, фіксує evidence і планує review.
- Published item має knowledge/outcome mapping, accepted answer policy, feedback, accessibility fields і всі required media references.
- Published content/assets versioned; rollback не втрачає progress і historical evidence.
- При повністю вимкненому AI користувач може пройти published lessons, Review, collections, audio і зберегти progress.

2. Межі версії v1

2.1. У межах v1

- Responsive Web/PWA із touch-first mobile UX і desktop UX.
- Український UI/explanations; English en-GB/en-US та Čeština cs-CZ.
- General English і General Czech з curriculum architecture A1-C1; content публікується module-by-module тільки після QA.
- Home, Review, Explore/Course, Lesson Player, Module Recap, Profile/Settings, Personal Collections, Progress.
- Content-driven Lesson Engine, two-level Learner/Mastery Model, Review Engine, pronunciation/audio та Media Pipeline.
- Admin/CMS для structured authoring, validation, approvals, versioning, publish/rollback, audit і AI settings.
- Optional AI capabilities for Learning та Administration через provider-agnostic AI Layer; feature-flagged і non-essential для core.
- Architecture support for professional tracks; Digital Management/PPC - перший planned track, насамперед English.

2.2. Не входить до v1 як обов’язкова core-функція

- PDF/textbook import.
- Вимога до learner самостійно шукати, завантажувати або створювати required lesson media.
- Live image/audio generation як dependency кожної звичайної вправи.
- Native iOS/Android apps, доки PWA не покаже measured need.
- Social network, marketplace, live tutors, official CEFR certification.

- Копіювання Babbel або іншої платформи: lesson text, assets, audio, sequence, branding, character design або recognizable trade dress.
- Payments/subscriptions до окремого commercial decision.
- Автономна AI-публікація curriculum, AI delete/role/billing operations або high-stakes certification decisions.
- Обов’язковий one-folder HTML prototype як stage: reference implementation одразу будується на production-oriented schemas/contracts.

3. Користувачі, ролі та права

Роль	Основні можливості	Обмеження
Learner	Навчання, Review, AI learner features if enabled, speaking, collections, profile/data controls.	Не змінює published curriculum/system assets.
Content editor	Lesson briefs, activities, notes, media requirements, AI-assisted drafts.	Не publish без review gates.
Linguistic reviewer	Language, variety, pronunciation, translation, register, accepted variants.	Не змінює production code.
Pedagogical reviewer	CEFR, sequence, cognitive load, distractors, assessment, AI pedagogical safety.	Не обходить schema/QA gates.
Administrator	Roles, feature flags, AI modes/providers/quotas, publish/rollback, audit, recovery.	Небезпечні дії audited/MFA; human authority retained.
Automated worker / AI service	Draft/generation/validation/jobs, allowed low-risk AI Auto actions.	Не publish/rollback/delete account/change roles/billing; service permissions мінімальні.

ID	Вимога	Пріоритет	Критерій перевірки
ROLE-001	RBAC MUST відокремлювати learner/editor/reviewer/admin/service operations.	MUST	Negative permission matrix.
ROLE-002	Publish, rollback, delete account, role change та AI budget/provider change створюють audit event.	MUST	actor/action/target/time/result; no secrets.
ROLE-003	Automated generation/AI не переводить draft у published напряму.	MUST	Workflow/API denies transition.
ROLE-004	AI Auto працює тільки за allowlist capabilities і не успадковує blanket admin authority.	MUST	Unknown action denied by policy engine.

4. Мови, локалізація та варіанти

Сфера	Код/варіант	Правило v1
Інтерфейс / explanations	uk-UA	UI strings via i18n keys; instructional content separate.
English	en-GB / en-US	en-GB default. en-US meaningful differences as variant note/accepted variant/asset.
Čeština	cs-CZ	Standard modern Czech; diacritics, morphology, register and pronunciation metadata.
Future languages	e.g. pl-PL, de-DE	Not in v1 launch content; schemas and engines already language-scoped.

ID	Вимога	Пріоритет	Критерій перевірки
LANG-001	Onboarding/settings expose real English / Čeština selector.	MUST	Both open real catalog/lesson flow.
LANG-002	Progress, mastery, review, collections, audio prefs and course position isolated by language profile.	MUST	Switch leaves other profile unchanged.
LANG-003	English content has explicit en-GB/en-US/common variety scope.	MUST	Validator rejects missing scope where applicable.
LANG-004	English/Czech text is linted for accidental Cyrillic characters; Czech diacritics preserved.	MUST	Script lint fixtures.

ID	Вимога	Пріоритет	Критерій перевірки
LANG-005	UI locale and target-language instructional data are independent.	MUST	Locale switch does not change target content.
LANG-006	Dates stored UTC; review/reminders presented in user timezone.	SHOULD	DST/day-boundary tests.

ЧАСТИНА B. ПЕДАГОГІКА, CURRICULUM І КОНТЕНТ

5. Педагогічна модель і CEFR

Curriculum спирається на CEFR і action-oriented learning. Рівень A1-C1 є структурою outcomes, а не автоматичною сертифікацією. Evidence має показувати, що learner може виконувати communicative action, а не лише завершив screens.

Нормативний learning flow
Context -> Exposure -> Noticing -> Explanation -> Recognition -> Guided retrieval -> Independent retrieval -> Listening/Reading -> Speaking/Writing -> New context -> Spaced review -> Higher-level reuse

- Scaffolding: support decreases as evidence grows.
- Spiral curriculum: important knowledge returns at higher levels with new function/complexity.
- Retrieval practice: learner actively recalls instead of rereading.
- Interleaving: Review mixes related knowledge without chaotic random jumps.
- Contextual recycling: old chunks/grammar reappear in new communicative situations.
- Guided discovery: examples and interactive gaps may precede concise rule.
- Productive transfer: where level permits, lessons lead toward speaking/writing/integrated task.
- Metacognitive strategy notes teach gist listening, repair, inference and efficient review.

ID	Вимога	Пріоритет	Критерій перевірки
PED-001	Each lesson has 1-3 verifiable learning outcomes.	MUST	outcome_refs + final task trace.
PED-002	New material does not immediately require unsupported free production.	MUST	Scaffolded path review.
PED-003	Review changes cognitive operation or context instead of duplicating same item endlessly.	MUST	Near-duplicate policy.
PED-004	CEFR claims map to outcomes/evidence, not lesson count.	MUST	Coverage/evidence report; no official certificate claim.

6. Curriculum і knowledge graph

Language -> CEFR Level -> Course/Track -> Module -> Lesson -> Activity -> Item. Curriculum задає ordered learning path; knowledge graph задає reusable knowledge, prerequisites, contrasts, components, variants, reuse і co-occurrence. PostgreSQL relational edges є baseline; окрема graph database не потрібна без measured need.

6.1. Типи knowledge items

Тип	Приклад	Що відстежується
Lexeme / sense	bank = financial institution	meaning, recall, spelling, listening, production, context
Chunk / collocation	make a decision	whole-chunk production, context, register
Grammar concept	past simple question	form, meaning, use, error patterns
Communicative function	ask for clarification	appropriateness, interaction strategy
Phoneme / feature	English /θ/, Czech ř	perception, pronunciation facet
Orthographic pattern	Czech vowel length	spelling, dictation, recognition
Culture / usage note	British vs American term	recognition/contextual choice; not separate grammar score by default

6.2. Основні зв’язки

- prerequisite_of
- component_of
- contrasts_with
- variant_of
- reused_in
- co_occurs_with

ID	Вимога	Пріоритет	Критерій перевірки
CURR-001	Each language has separate A1-C1 curriculum with shared schemas/engines.	MUST	EN/CS fixtures without engine fork.
CURR-002	Each assessable item maps to knowledge item(s) and evidence targets.	MUST	Publish validation rejects missing map.
CURR-003	Graph edges have type/direction/version/review state.	MUST	Admin can inspect prerequisite path.
CURR-004	Prerequisites guide recommendation but optional Level Test or manual level switch does not hard-block access.	MUST	UI warns gaps, allows access.
CURR-005	Coverage report exposes outcome/knowledge/activity coverage and duplicate gaps.	SHOULD	Editor report before publish.

7. Структура уроку

Real-life context -> New input -> Vocabulary/Sentence intro -> Interactive explanation -> Controlled practice -> Retrieval -> Listening -> Dialogue -> Speaking/Pronunciation -> Integrated task -> Result -> Mistake remediation. Не кожен lesson має всі blocks; context, active retrieval і visible result є обов’язковими.

Блок	Мета	Типові activity
Context	Навіщо потрібна мова	scene, dialogue preview, image, short audio
Input/noticing	Побачити/почути target forms	sentence cards, highlighting, audio
Explanation	Build concise rule/usage note	interactive grammar, staged reveal, contrast
Controlled practice	Reduce cognitive load	choice, matching, gap fill, builder
Retrieval	Recall without full answer	typing, ordering, L1->L2, listening recall
Communication	Use in meaningful context	dialogue chain, role-play, writing, speaking
Result/remediation	Show outcome and repair errors	mistake queue, feedback, next review

8. Lesson Engine та activity contract

Lesson Engine reads versioned structured content. Educational text, answers, feedback and media references are not hardcoded in React/HTML renderers. New activity type enters through registry + schema + evaluator/completion contract + fixtures.

8.1. Core activity types

Vocabulary/Sentence/Scene intro; Multiple choice; Image choice; Matching; Classification; Gap fill; Word builder; Sentence ordering; Full typing; Translation; Listening gist/detail; Reading comprehension; Dictation; Transcript reveal; Dialogue completion/chain; Role-play; Read aloud; Listen & repeat; Interactive grammar; Pronunciation contrast; Writing; Integrated task; Module recap; optional Level Test.

8.2. Мінімальний activity schema

id, schema_version, type, language, variety, CEFR level, instruction_key, prompt, stimulus, response_schema, accepted_answers/policy, distractors, feedback, hint/reveal policy, knowledge_refs, evidence_map, media_refs, accessibility, randomization, analytics_tags, feature requirements.

ID	Вимога	Пріоритет	Критерій перевірки
LESS-001	One renderer works with all items of its activity type.	MUST	English/Czech contract fixtures.

ID	Вимога	Пріоритет	Критерій перевірки
LESS-002	Accepted-answer normalization is explicit and language-aware.	MUST	Case/whitespace/punctuation/ diacritics/variants fixtures.
LESS-003	Distractor cannot accidentally be another correct answer.	MUST	Automated + linguistic review.
LESS-004	Randomization preserves correctness and is reproducible for debugging.	SHOULD	Seed/order retained.
LESS-005	Instructions, answer feedback and audio controls keyboard-accessible.	MUST	Keyboard E2E core types.
LESS-006	Unsupported schema returns recoverable state, not blank screen.	MUST	Renderer error fixture.

9. Exercise state machine і remediation

not_started -> active -> submitted -> correct | incorrect | assisted -> remediation -> completed -> scheduled_review

- correct = independent answer; evidence depends on assistance/difficulty/activity.
- incorrect = brief objective-specific feedback + retry/simpler scaffold.
- assisted = reveal/answer-giving hint; not independent mastery.
- remediation = after main lesson, unresolved scored mistakes are revisited until correct/reveal or explicit exit with persistence.
- An item corrected after a lapse still gets earlier review than a clean first-try retrieval.

ID	Вимога	Пріоритет	Критерій перевірки
STATE-001	Attempt transition is valid and idempotent.	MUST	Duplicate attempt_id no duplicate score/effects.
STATE-002	Reveal is not independent correct.	MUST	assisted evidence class.
STATE-003	Lesson result exposes unresolved scored mistakes and remediation/exit path.	MUST	E2E mistake loop.
STATE-004	Feedback is concise, specific and language-aware.	MUST	Error tag/cause review.

10. Learner Model, mastery та evidence

Two-level mastery model
1) Six global skill projections: Vocabulary, Grammar, Listening, Speaking, Writing, Pronunciation. 2) Per-knowledge-item facets: recognition, meaning, production, spelling, listening_recognition, pronunciation, context_use and reviewed language-specific facets such as collocation, morphology, word_order, register. A single overall score MUST NOT hide missing facet coverage.

Global skill projection	Evidence examples	Typical item facets
Vocabulary	lexical recall/meaning/chunks across items	recognition, meaning, production, spelling, context_use
Grammar	form, meaning, agreement, morphology, word order	morphology, word_order, context_use, register where relevant
Listening	audio-only gist/detail/identification	listening_recognition, meaning
Speaking	usable learner audio in dialogue/production	production, context_use, interaction facets
Writing	builder/typing/free written response	production, spelling, context_use
Pronunciation	usable audio for named target features	pronunciation, intelligibility target

ID	Вимога	Пріоритет	Критерій перевірки
MAST-001	Mastery state stores estimate, uncertainty, evidence_count, last_evidence_at, lapse_count and model_version.	MUST	Data/API contract.

ID	Вимога	Пріоритет	Критерій перевірки
MAST-002	Evidence weight depends on activity/evidence type, hints/reveal, attempts, latency/freshness and difficulty.	MUST	Deterministic tests.
MAST-003	Multiple surfaces mapping to same knowledge item update same facet/mastery projection.	MUST	No duplicate mastery rows.
MAST-004	Model changes do not rewrite historical attempts/evidence.	MUST	Versioned recompute.
MAST-005	CEFR/course progress is aggregate view, not sole truth.	MUST	Reset/transition preserves evidence unless explicit.
MAST-006	Global skills and item facets are distinct layers and may not be collapsed into one dimension key.	MUST	Schema + projection tests.
MAST-007	AI feedback does not change mastery unless validated evaluator/rubric/activity contract authorizes evidence.	MUST	AI/offline comparison produces same core mastery.

11. Review i spaced repetition

- Recommended today; All vocabulary; Mistakes; Words; Phrases/chunks; Topics; Personal collections.
- Skill modes: Vocabulary, Listening, Speaking, Writing, Grammar, Pronunciation.
- Inputs: correctness, attempts, reveal, hints, response time, item difficulty, time since retrieval, error type, mastery/facet state.
- Session: due first, then weak; limited new/recycle; interleaving; spacing between near-duplicates.
- Skip is not success and does not postpone due date like correct retrieval.
- Topic selection and personal collections can constrain what is reviewed, while scheduler still prioritizes weak/due items.

ID	Вимога	Пріоритет	Критерій перевірки
REV-001	Due schedule is isolated by language profile and evidence target/facet.	MUST	No cross-language item.
REV-002	Clean correct, hinted correct, incorrect and reveal have different scheduling effects.	MUST	Scheduler unit tests.
REV-003	Session builder deduplicates same knowledge/surface.	MUST	Knowledge refs + normalized hash.
REV-004	Scheduler is versioned and may be shadow/A-B tested.	SHOULD	scheduler_version analytics.
REV-005	Review session has bounded size and ETA.	SHOULD	UI/count policy cap.

12. Speaking, pronunciation, listening та audio

- Read aloud; Listen & Repeat; Role-play; dialogue playback by speaker.
- Normal + slow playback; slow mode preserves pitch or uses reviewed slow asset.
- Voice assets scoped en-GB/en-US/cs-CZ; multi-speaker dialogues bind voice_id/avatar/current audio state.
- Microphone button begins recording only after explicit user action; while recording it visually pulses unless reduced-motion is active.
- For usable/high-confidence speech, learner-facing result is simple correct or try again/incorrect; raw numeric score is not required in UI.
- On correct, Next becomes available or Auto-advance may proceed according to setting. On incorrect, concise feedback and retry/continue behavior follow activity policy.
- Mic denial, unsupported device, timeout, provider failure or low-confidence recognition => speech_unavailable; neutral, non-blocking, no negative Speaking/Pronunciation evidence.
- Raw microphone audio is transient by default and not stored in logs/analytics/object storage without explicit separate policy.

ID	Вимога	Пріоритет	Критерій перевірки
AUD-001	Required audio has normal asset; slow via quality transform or approved asset.	MUST	Playback/duration/language QA.
AUD-002	Microphone permission only after explicit action.	MUST	Fresh browser E2E.
AUD-003	Raw recording transient by default.	MUST	Storage/log audit.
AUD-004	Pronunciation threshold/normalization configured per language/variant.	MUST	EN/US/CS fixtures.
AUD-005	Audio has transcript/text alternative where compatible with objective.	MUST	Accessibility review.
AUD-006	TTS/STT/speech integrations use provider adapters.	SHOULD	Adapter substitution.
AUD-007	Usable speech may show binary correct/try again; unavailable/low confidence remains neutral and cannot hard-block learning.	MUST	Speech state E2E + mastery assertion.

13. Personal collections

- Create, rename, edit, archive/delete; add words, phrases, chunks, questions and useful sentences.
- Play all audio and Review only this collection.
- Collection entry links to knowledge/content entity when available; custom text stored separately.
- Collection language is explicit; mixed-language collection is not created accidentally.

ID	Вимога	Пріоритет	Критерій перевірки
COLL-001	Collections isolated by user + target language.	MUST	API ownership/language tests.
COLL-002	Deleting collection never deletes global knowledge/mastery history.	MUST	Referential integrity.
COLL-003	Custom entry supports normalized text, optional translation/note and audio status.	SHOULD	UI handles missing audio.

14. Content production, CMS i publication workflow

Course map -> Lesson brief -> Structured draft -> Automated validation -> Linguistic review -> Pedagogical review -> Media generation -> Integrated QA -> Preview -> Publish -> Monitor -> Revise

> **Роль системи**
> Lingvessa tooling/assistant may prepare curriculum drafts, exercises, explanations, dialogues, images, audio scripts and QA suggestions. Product owner does not need to manually search for photos or write all assets. AI output remains draft/proposal until required automatic/editorial gates pass.

14.1. Статуси контенту

idea -> planned -> draft -> validation_failed | review_ready -> linguistic_review -> pedagogical_review -> media_ready -> qa_ready -> approved -> published -> superseded | archived

14.2. Автоматичні перевірки перед publish

- Schema/version/required fields/IDs/refs/language/variant/CEFR.
- Accidental Cyrillic in English/Czech, missing Czech diacritics where required, broken markup, duplicate IDs.
- Unambiguous correct response, distractors, feedback/hint/reveal policy.
- Knowledge/outcome/evidence mapping, required media, alt/transcript, pronunciation metadata.
- Provenance/rights, generation records, approvals, no placeholder content.
- Preview in core viewports and keyboard navigation for new activity types.

ID	Вимога	Пріоритет	Критерій перевірки
CONT-001	Published content is immutable snapshot with content_version.	MUST	Edit creates new version.
CONT-002	CMS shows diff and supports rollback to approved version.	MUST	Staging rollback.
CONT-003	Publish blocked without mandatory linguistic/pedagogical/technical gates.	MUST	Workflow negative tests.
CONT-004	Content migration cannot silently lose progress/mastery.	MUST	Impact mapping report.
CONT-005	Bulk generation has rate limits, retry policy, cost telemetry and deterministic job IDs.	SHOULD	No duplicate assets/items.

15. Media requirement і generation pipeline

Curriculum -> Lesson/Item -> Media requirement -> Prompt/spec -> Generation job -> Automated QA -> Human/media review -> Object storage -> asset_id + asset_version -> Lesson binding -> Publish

Основний режим - попередня генерація та збереження assets до публікації. Production lesson references stable asset_id + version, not provider URL or ad-hoc filename. AI can orchestrate this pipeline, but learner does not generate required media during ordinary lesson runtime.

Metadata	Мінімальний зміст
Identity	asset_id, asset_version, kind, status, checksum
Scope	language, variety, course, lesson/item purpose, audience
Generation	provider/model, prompt/spec hash, job id, created_at, parent/reference
Quality	dimensions/duration/format/loudness/visual checks/review state
Accessibility	alt text, transcript, speaker labels, captions where needed
Rights/provenance	origin, usage basis, restrictions, reviewer, retention
Consistency	character/style bible version, voice_id, pronunciation lexicon version

ID	Вимога	Пріоритет	Критерій перевірки
MED-001	Learner does not provide media required for standard course.	MUST	Published lesson complete with system assets.
MED-002	Broken/missing required asset blocks publish.	MUST	Preflight existence/checksum/format/CDN.
MED-003	Recurring characters/voices use versioned consistency profiles.	MUST	Series assets share profile IDs.
MED-004	Provider URL is never production lesson dependency.	MUST	Platform-controlled storage after ingest.
MED-005	Regeneration creates new asset_version/checksum.	MUST	Rollback cache-safe.
MED-006	Real-person voice/face cloning requires separate legal basis + explicit consent.	MUST	Media gate blocks unknown identity provenance.

16. AI capability layer

Lingvessa has one provider-agnostic AI Layer for learner-facing and admin-facing capabilities. AI improves explanation, practice, personalization and operations, but cannot replace curriculum, deterministic state machine, published content governance or learner evidence model.

Core-without-AI invariant
If every AI provider is disabled, rate-limited or offline, learner MUST still be able to log in, open published lessons, complete deterministic exercises, receive normal feedback, save progress, use Review, collections, audio assets and resume learning.

16.1. AI for Learning

Capability	Use	Authority
Пояснити інакше	Grounded personalized explanation	Advisory; approved curriculum/knowledge is authority.
Adaptive hints	Context-sensitive hints within hint/reveal policy	Cannot silently give answer beyond activity policy.
Extra practice	Generate extra examples/items around weak facets	Practice-only unless deterministic validated template.
Writing feedback	Grammar/lexis/register feedback	Advisory by default; mastery only with validated rubric.
AI conversation/role-play	Text conversation; voice through existing STT/TTS adapters	Optional/feature-flagged; evidence rules separate.
Error-focused remediation	Explain recurring errors and propose practice	Cannot overwrite immutable evidence/history.

16.2. AI for Administration

Режим	Призначення	Межа
Manual	AI off or suggestions requested manually	Human performs all changes.
AI Assist	Default recommended admin mode	AI drafts/suggests; human reviews/applies/publishes.
AI Auto	Explicit opt-in per capability	Only allowlisted low-risk draft/admin actions with diff + audit.

16.3. Cost/provider policy

- Provider-neutral adapters; no domain schema depends on provider SDK/data shape.
- Free/local-first is the startup strategy, but provider eligibility depends on privacy, terms, quality and quota - not only price.
- Paid inference is disabled by default; monthly paid AI budget = 0 until product owner explicitly approves change.
- No free tier is assumed permanent; provider can be switched without changing learner/admin feature contracts.
- AI provider failure triggers fallback or non-AI path; no hidden runtime dependency.

ID	Вимога	Пріоритет	Критерій перевірки
AI-001	AI is optional for core learner path.	MUST	Provider-offline EN/CS reference E2E.
AI-002	AI output is non-authoritative until validated by contract/evaluator/human gate.	MUST	AI cannot directly set correct/mastery/published/role.
AI-003	AI behavior is traceable to provider/model/prompt/policy versions.	MUST	Audit/support reproduction.
AI-004	Paid AI is blocked under default configuration with budget 0.	MUST	Cost guard tests.
AI-005	AI Assist is admin default; AI Auto is allowlist-based and draft/low-risk only.	MUST	Permission matrix.
AI-006	AI service identity cannot publish/rollback/delete account/change role/billing.	MUST	Authorization negative tests.
AI-007	Learner AI grounds on approved curriculum/knowledge/language profile and cannot treat model memory as curricular authority.	MUST	Grounding/eval fixtures.
AI-008	AI-generated course media enters only through existing Media Pipeline and publish gates.	MUST	No direct provider asset binding in published lesson.

ЧАСТИНА C. UX ТА ДОСТУПНІСТЬ

17. Інформаційна архітектура та основні flows

Розділ	Призначення	Ключові елементи
Onboarding	Target language, starting level/goal	English/Čeština; optional placement; AI/privacy choices where enabled
Home	Next useful step	Continue, daily review, streak/progress, active language
Review	Due/weak/mistake practice	recommended, topics, collections, skill modes
Explore / Course	A1-C1 and tracks	module map, prerequisites, status, repeat
Lesson Player	One activity at a time	instruction, context, answer, audio, AI help optional, progress, exit
Result	Evidence without punishment	mistakes, remediation, relevant skill indicators, next review
Progress	Longitudinal profile	global skills, coverage, trends; detailed facets on demand
Profile / Settings	Languages, variants, reminders, data, AI	reset/export/delete; AI mode/provider options for admin; learner AI toggles

ID	Вимога	Пріоритет	Критерій перевірки
UX-001	Active target language visible in core navigation.	MUST	No accidental cross-language lesson.
UX-002	Mobile bottom nav/touch-first; desktop wider/top-nav layout.	MUST	Breakpoint QA.
UX-003	Each lesson state has Back/Close/Continue/recovery paths.	MUST	E2E correct/incorrect/reveal/network/ speech.
UX-004	Auto-advance is a setting and only applies from allowed completion state.	MUST	Toggle/state test.
UX-005	Reset lesson/module/language scope has preview + confirmation.	MUST	Scope shown before mutation.
UX-006	Lesson Result need not show all six skills every time; full profile belongs in Progress.	SHOULD	Result remains concise; Progress exposes profile.

18. Responsive design та accessibility

Target: WCAG 2.2 AA for learner-facing core flows, independent of legal applicability in launch country.

- Keyboard navigation, visible focus, logical tab order, names/roles/states.
- 200% zoom/reflow; core content from 320 CSS px without horizontal scroll.
- Primary touch targets >= 44x44 CSS px; reduced motion; no information by colour alone.
- Alt text/decorative markers; transcripts/captions; non-drag alternatives.
- Mic pulse respects prefers-reduced-motion and is never sole recording state signal.
- AI learner/admin UI must expose what is AI-generated where policy requires and have accessible non-AI fallback for core learning.

ID	Вимога	Пріоритет	Критерій перевірки
A11Y-001	Core learner flows pass automated + manual WCAG 2.2 AA review.	MUST	Automated + manual evidence.
A11Y-002	Screen reader announces instruction, answer state, error, progress once in useful order.	MUST	Desktop/mobile SR scenarios.
A11Y-003	Animation/pulsing microphone respects reduced motion.	MUST	Text/icon state remains clear.
A11Y-004	Language attributes set correctly for uk/en-GB/en-US/cs.	MUST	DOM/lang audit.

19. Profile, settings і notifications

- Interface language; target language; English variety; goals; active course/track.
- Auto-advance; normal/slow audio; captions/transcript; microphone controls.
- Learner AI enable/disable and privacy-sensitive feature toggles where AI is available.
- Reminder cadence, quiet hours, timezone; notifications only after opt-in.
- Reset lesson/module/language progress; export; delete account.
- Switching target language does not erase progress/mastery/review/collections of other language.

ID	Вимога	Пріоритет	Критерій перевірки
SET-001	Settings have server-side source of truth and safe local cache.	MUST	Restore on another device.
SET-002	Reminder respects opt-in, timezone and quiet hours.	MUST	DST/off tests.
SET-003	Delete account has confirmation/grace flow per data policy.	MUST	Deletion job/audit.
SET-004	AI settings are capability-scoped; disabling AI cannot disable core lessons/review/progress.	MUST	Provider-disabled E2E.

ЧАСТИНА D. ТЕХНІЧНА АРХІТЕКТУРА ТА ДАНІ

20. Архітектурні принципи

- Content-driven: structured content/schema separate from rendering code.
- Modular monolith first + workers for long/cost-heavy jobs; microservices only after measured need.
- Provider-neutral: image, TTS, STT, AI, email/push, storage via adapters/internal contracts.
- Language-scoped by design in schema, storage, cache keys, analytics and tests.
- Version everything that affects learning: content, assets, schema, mastery/scheduler, prompt/style guide, pronunciation lexicon, AI policy.
- Privacy/accessibility by design, not end-of-project patches.
- Private source control: code/config/schemas/scripts in private Git repository; no secrets in repository.
- Hosting is replaceable infrastructure; domain model/contracts must not depend on a specific free hosting vendor.

21. Рекомендований технологічний baseline

Не незмінний контракт із vendor
Domain model, content schemas and acceptance criteria have priority over hosting choice. Exact framework/provider versions are pinned in repository after compatibility checks.

Шар	Baseline	Межа
Web UI	TypeScript, React, Next.js App Router, PWA	Responsive learner/admin UI; original design system.
Application API	TypeScript/Node.js modular monolith	Lesson/Review/Content/Media/Identity/AI policy modules.
Database	PostgreSQL	Source of truth; relational integrity; JSONB only for versioned payloads/metadata.
Queue/cache	Redis-compatible as needed	Rate limits/jobs/cache; never source of truth.
Media	S3-compatible object storage + CDN	Stable asset IDs/versions/checksums.
Workers	Node.js workers + media tooling	generation/transcoding/validation/notifications/AI jobs.
Validation	JSON Schema + typed runtime validation	Shared by authoring/API/tests/preflight.
Testing	Unit/integration + Playwright + a11y/visual	Core states, languages, provider fallback.
Delivery	OCI containers, CI/CD, managed stores, secrets manager	Repeatable environments and portable deployment.
Repository	Private Git repository; GitHub Private current development default	All source code/versioned schemas/scripts; media bytes excluded when large.
AI	Provider-agnostic AI adapters; free/local-first; paid optional later	No AI dependency in core; budget controls/fallback.

ID	Вимога	Пріоритет	Критерій перевірки
TECH-001	Temporary prototype code does not become source of truth without explicit migration review.	MUST	Production uses schemas/tests/persistence.
TECH-002	Domain modules do not import provider SDK outside adapter layer.	MUST	Architecture lint.
TECH-003	Dependencies lockfile + vulnerability checks + controlled upgrades.	MUST	CI reproducible build.
TECH-004	Secrets never enter client bundle/source control/logs.	MUST	Secret scan + bundle inspection.
TECH-005	Feature flags can scope language/course/activity/provider /AI capability.	SHOULD	Disable without breaking existing sessions.

22. Доменна модель даних

Entity	Призначення	Ключові правила
User / Account	Identity, status, consent refs	No direct course progress field.
LanguageProfile	Independent English/Czech state	user+language; variety, goal, current position.
Course/Level/Module/Lesson	Curriculum hierarchy	Versioned, language-scoped.
Activity/Item	Renderable structured task	schema_version, knowledge_refs, media_refs, evidence map.
KnowledgeItem/Edge	Knowledge graph	language/variant/type/dependencies.
Attempt/EvidenceEvent	Immutable learning event	user/profile/content/evaluator versions.
MasteryState	Item-facet estimate/projection state	profile+knowledge+facet/model version.
SkillProjection	Global 6-skill aggregate	Derived from validated evidence/facets.
ReviewSchedule	due_at/scheduler state	knowledge + facet/evidence target + version.
Collection/Entry	Personal sets	owner + language; curated or custom.
Asset/AssetVersion	Media identity/bytes versions	checksum/storage/provenance/language/QA.
ContentVersion/Publication	Immutable release snapshot	approvals/manifest/rollback pointer.
AIConfig/AIJob/AIAudit	AI routing/mode/policy/output metadata	No hidden CoT; prompt/model/policy versions, cost/decision trail.
Consent/AuditEvent	Privacy and privileged actions	Append-only where required.

ID	Вимога	Пріоритет	Критерій перевірки
DATA-001	Primary entities use stable opaque IDs, not filenames/titles/indexes.	MUST	Rename/reorder keeps refs/progress.
DATA-002	Attempt/Evidence immutable except documented redaction/deletion.	MUST	No generic update API.
DATA-003	User-scoped reads/writes check ownership server-side.	MUST	IDOR tests.
DATA-004	Content payload has schema_version and migration path.	MUST	Old fixtures render/migrate.
DATA-005	Analytics/AI logs are not source of truth for progress.	MUST	Outage does not affect completion.
DATA-006	Mastery storage distinguishes item facets from global skill projections.	MUST	Schema prevents dimension collapse.

23. API, jobs та синхронізація

- Versioned API contracts for catalog, lessons, attempts, review sessions, profile/settings, collections, media, admin/content and AI capabilities.

- Attempt submit uses client-generated attempt_id/idempotency and returns authoritative result/effects.
- Long-running generation, transcoding, AI tasks, bulk validation, export/delete and notifications run as jobs with status/retry/timeout/dead-letter.
- Offline/unstable connection queues safe unsent attempts; reconnect preserves order/idempotency.
- Provider callbacks verify signature/job identity where applicable.
- AI requests have capability/policy/budget checks before provider call; provider failure returns typed fallback.

ID	Вимога	Пріоритет	Критерій перевірки
API-001	API errors use stable code, safe message and correlation id.	MUST	UI distinguishes validation/auth/conflict/rate/service unavailable.
API-002	Attempt/destructive admin operations are idempotent or conflict-safe.	MUST	Retry tests.
API-003	Server authoritative for answer evaluation, due_at and privileged transitions.	MUST	Tampering has no effect.
API-004	Jobs have observable lifecycle and bounded retries.	MUST	Failed/stuck visible; no infinite retry.
API-005	Rate limiting separates ordinary learner actions from speech/generation/AI cost-heavy actions.	SHOULD	Separate quotas/load tests.
API-006	AI API cannot directly authoritatively mutate published content, mastery or roles.	MUST	Authorization/evidence tests.

24. Repository, storage, deployment і versioning

- Relational metadata in PostgreSQL; large media bytes in object storage; delivery through CDN.
- Published manifest pins content_version + asset_version; preview uses draft manifest.
- Cache keys include version/checksum; replacing bytes under same checksum/version is forbidden.
- Backups cover DB, publication manifests and critical asset metadata; object storage lifecycle/versioning protects referenced assets.
- All source code, schemas, migrations, content tooling, Docker/IaC/config templates and docs are versioned in private Git repository.
- GitHub Private is the current development repository default; secrets use environment/secret stores, never committed files.
- Hosting/provider topology is selected via ADR and can change without rewriting core architecture. Development/test may use free tiers, but production constraints are decided separately.

ЧАСТИНА E. ДОВІРА, ЯКІСТЬ І ЕКСПЛУАТАЦІЯ

25. Security, privacy та захист даних

Security baseline aligns at least with OWASP ASVS 5.0 and current web practices. For public launch, privacy/jurisdiction/minor/provider decisions are reviewed for actual countries and data flows; this document is not a legal opinion.

- TLS, encryption at rest for managed stores, least privilege, MFA for privileged roles.
- Secure sessions/cookies, CSRF/XSS/injection protections, server-side authorization, abuse/rate controls.
- Data minimization: no raw microphone audio, full free responses, device identifiers or personal AI context without defined purpose.
- Export/correction/deletion and retention schedules for account, learning data, logs, backups, AI traces and media metadata.
- Before minors: age gate/parental consent/jurisdiction decision; no universal guessed threshold.
- Third-party providers reviewed for data-use/training/regions/subprocessors; free availability never overrides privacy policy.
- AI outbound payloads use allowlists; prompt injection/untrusted content treated as security input.

ID	Вимога	Пріоритет	Критерій перевірки
SEC-001	Server-side authorization on every protected object/action.	MUST	ASVS-oriented IDOR/privilege tests.
SEC-002	Sensitive values redacted from logs/analytics/error/AI traces.	MUST	Automated canary scan.

ID	Вимога	Пріоритет	Критерій перевірки
SEC-003	Self-service export/delete flow per approved policy.	MUST	E2E lifecycle.
SEC-004	Production admin protected by MFA + least privilege.	MUST	No MFA -> no privileged command.
SEC-005	Threat model, processor inventory and retention schedule before public launch.	MUST	Release gate artifacts.
SEC-006	Age/minor flow fixed only after jurisdiction/legal review.	MUST	Registration gating.
SEC-007	AI provider routing respects data classification, approved purpose and retention policy.	MUST	Privacy route tests.

26. Originality, copyright i AI transparency

- Own curriculum map, lesson text, explanations, examples, dialogues, exercises, images, audio, branding, icons and visual system.
- No copying Babbel/other product by screenshot, text, sequence, characters, sounds or recognizable combined UI decisions.
- General methods (multiple choice, SRS, progress bar) are implemented independently.
- Every generated/imported asset has provenance/usage basis; third-party reference only when rights allow.
- AI-generated content/media follows same originality, rights and review gates as human-authored content.
- AI transparency obligations are assessed per flow/jurisdiction; do not make blanket compliance claim without launch-specific review.
- No hidden chain-of-thought is stored or exposed as product data; only final outputs and necessary structured policy/trace metadata.

ID	Вимога	Пріоритет	Критерій перевірки
LEGAL-001	Published item/asset has provenance/rights status.	MUST	Unknown/restricted blocks publish.
LEGAL-002	Originality review covers text, branding, layout patterns, characters and audio cues.	MUST	Documented audit before public launch.
LEGAL-003	AI transparency evaluated per flow/jurisdiction.	MUST	Decision log.
LEGAL-004	Provider terms and data-use settings recorded for generation/speech/AI vendors.	MUST	Vendor register version/date/purpose.

27. Non-functional requirements

ID	Вимога	Пріоритет	Критерій перевірки
NFR-PERF-001	Core pages target p75 LCP <=2.5s, INP <=200ms, CLS <=0.1 on supported real-user segments.	SHOULD	RUM/lab regression.
NFR-PERF-002	Normal cached API read p95 <=400ms; attempt submit <=700ms excluding external speech/AI/media jobs.	SHOULD	Load/production metrics.
NFR-REL-001	Accepted attempt survives retry and analytics/AI outage.	MUST	Fault injection.
NFR-REL-002	Initial learner core availability target 99.5% monthly excluding announced maintenance.	SHOULD	SLO dashboard.
NFR-DR-001	Initial production RPO <=24h, RTO <=8h for database/critical manifests.	MUST	Restore drill.
NFR-COMP-001	Support current major browsers and two recent major versions where practical.	SHOULD	Browser matrix.

ID	Вимога	Пріоритет	Критерій перевірки
NFR-SCALE-001	Web/API can scale stateless; durable state not process-memory only.	MUST	Multi-instance test.
NFR-LOC-001	Unicode/Czech diacritics/Ukrainian round-trip preserved.	MUST	Normalization fixtures.
NFR-AI-001	AI provider outage/limit must not lower core lesson availability target.	MUST	Provider-offline SLO/fault test.

28. Analytics та observability

Тип	Приклади	Правило
Learning analytics	lesson/activity/review/hint/reveal/speech fallback	Pseudonymous IDs; versions; no raw mic audio.
Product analytics	onboarding, language switch, return/drop-off	Consent/legal basis by launch; payload minimization.
Operational	latency, errors, queue depth, provider cost/failure	No answers/secrets.
AI quality	capability use, fallback, grounding/eval, cost/quota	Prompt/model/policy versions; no hidden CoT.
Audit	publish, rollback, role/delete/export, AI mode/provider/budget changes	Privileged append-only trail.

- Product metrics: lesson completion, review return, due backlog, retention, hint/reveal rate, mastery gain, speech fallback, AI optional-use/fallback.
- Content quality: ambiguity/error by item, unexpected valid alternatives, drop-off, media failures, AI suggestion acceptance/rejection.
- Analytics never mutates progress and may be unavailable without blocking lesson.

29. Testing, QA та release gates

Рівень	Що перевіряється	Gate
Schema/content lint	IDs/refs/scripts/answers/variants/media/ provenance/AI metadata	Every content/schema change
Unit	evaluation/state/mastery/facets/scheduler/AI policy/budget	Every code change
Integration	DB/auth/jobs/storage/provider adapters/privacy	Merge/release
E2E	onboarding, language switch, lesson states, review, settings, AI fallback, deletion	Release candidate
Visual/a11y	320px/tablet/desktop, zoom, long strings, keyboard/SR/motion	UI/core release
Linguistic	English variety, Czech morphology/diacritics, Ukrainian explanations, AI grounded output	Each lesson/module/AI prompt set
Pedagogical	outcomes, scaffolding, distractors, cognitive load, recap, AI help boundaries	Each lesson/module
Security/privacy	ASVS, threat model, secrets, provider routing, AI abuse/prompt injection	Public release
Restore/rollback	DB/content/media/config restore	Before production + drills

Definition of Done для published lesson
Schema valid; outcomes/knowledge/evidence mapping complete; answers unambiguous; feedback/hints reviewed; required assets approved; accessibility complete; language lint clean; linguistic/pedagogical/technical approvals present; preview mobile+desktop passed; content version published; analytics tags valid. Optional AI may enrich the lesson but must not be required for this DoD.

30. Ризики та пом’якшення

Ризик	Наслідок	Пом’якшення
English + Czech A1-C1 large scope	Uneven quality / long preparation	Schema-first, course maps, batch generation, lint, staged QA, measurable coverage.

Ризик	Наслідок	Пом’якшення
AI inconsistency/hallucination	Wrong explanations/practice	Grounding, schema/rubric, evals, human gates, AI optionality.
Speech false negative	Frustration / false mastery	Language calibration, confidence threshold, neutral unavailable fallback, simple correct/try again only when usable.
Provider dependency/cost	Price change/outage	Adapters, storage independence, free/local-first, budget guards, fallback.
Duplicate surfaces	Inflated mastery	Knowledge graph, hashes, facet mapping, deduplication.
Similarity to Babbel	IP/trade dress risk	Original visual/content system, provenance, originality audit.
Minors + voice/AI data	Privacy/legal risk	Transient audio, data classification, age/jurisdiction and provider review.
Overengineering	Slow first useful version	Modular monolith, production-oriented bilingual reference slice, no unnecessary microservices.
Free hosting/provider limits	Unstable development environment	Portable containers/PostgreSQL/S3 contracts; dev free tiers treated as replaceable, not product architecture.

ЧАСТИНА F. ПЛАН РЕАЛІЗАЦІЇ ТА ПРИЙМАННЯ

31. Етапи реалізації

Етап	Результат	Exit evidence
P0. Baseline sync	Approve synchronized specs, schemas, decision log, bilingual fixtures.	Formal v1.7 + aligned technical/curriculum/reference docs; AI v1.4; no hidden blocker.
Legacy descriptive stage 1 Repository + reference foundation	Private monorepo, CI, Web/API/DB skeleton, versioned schemas, bilingual reference data.	Clean checkout/build; EN+CS fixture validation; no temporary one-folder dependency.
Legacy descriptive stage 2 Learning core	LessonPlayer, registry, attempts, state, persistence, resume/offline queue.	English/Czech reference lesson E2E.
Legacy descriptive stage 3 Mastery + Review	Two-level mastery, schedules, mistakes, collections, reset semantics.	Deterministic facets/skills + language-isolated queues.
Legacy descriptive stage 4 Audio/Speech/Media	Assets, normal/slow, mic states, binary usable speech UX, provider fallback.	en-GB/en-US/cs-CZ flows; transient audio audit.
Legacy descriptive stage 5 Content/Admin system	CMS, approvals, generation jobs, versioned publishing/rollback, AI admin modes.	Draft-to-publish flow; AI cannot bypass gates.
Legacy descriptive stage 6 AI optional layer	Free/local-first adapters, explanation/hints/practice/writing/conversation flags, budget/privacy controls.	Provider-offline core E2E; paid budget 0; AI evals.
Legacy descriptive stage 7 Curriculum rollout	General English/Czech content A1-C1 module-by-module.	Coverage + editorial/media/a11y QA; no placeholder levels.
Legacy descriptive stage 8 Hardening/launch	Security/privacy/performance/backups/ observability/legal decisions.	Release gates + restore drill + public launch approvals.

Паралельний потік
After content/knowledge/activity schemas stabilize, course maps, lessons and media requirements may be authored in parallel. Bulk generation waits for validation, versioning and review workflow. AI does not need to be complete before core learning implementation starts.

32. Критерії приймання стартового продукту

ID	Вимога	Пріоритет	Критерій перевірки
ACC-001	English і Čeština available in onboarding/settings and complete core lesson flow.	MUST	New-account E2E both languages.

ID	Вимога	Пріоритет	Критерій перевірки
ACC-002	Progress/mastery/review/ collections fully isolated between languages.	MUST	Mixed data assertions.
ACC-003	Published educational content not hardcoded in UI components.	MUST	Fixture swap without code edit.
ACC-004	All required images/audio exist and bind via asset_id+version.	MUST	Manifest preflight/provider offline.
ACC-005	Mistakes/reveal update evidence/facets and next review correctly.	MUST	Deterministic state/mastery/scheduler tests.
ACC-006	Mobile/desktop core flows have no dead ends/clipping/horizontal overflow.	MUST	Visual/E2E matrix.
ACC-007	Core accessibility gate passes.	MUST	Automated + manual WCAG evidence.
ACC-008	Publish/rollback and asset rollback reproducible in staging.	MUST	Documented drill.
ACC-009	Security/privacy gates include audio, AI routing and account deletion.	MUST	Threat model + lifecycle checklist.
ACC-010	Originality audit finds no copied proprietary content/assets/branding/trade dress.	MUST	Review record.
ACC-011	Monitoring shows errors/latency/queue/provider failures/content versions/AI cost-fallback.	MUST	Fault injection.
ACC-012	Backup restore meets RPO/RTO.	MUST	Restore report.
ACC-013	Usable pronunciation speech shows correct/try again; low-confidence/provider failure neutral and non-blocking.	MUST	Speech fixtures and mastery assertions.
ACC-014	Two-level mastery exposes six global skills and separate item facets without cross-language leakage.	MUST	Schema/projection/review tests.
ACC-015	Core learning completes with all AI providers disabled and paid AI budget remains 0 by default.	MUST	Provider-offline E2E + budget guard.

33. Зафіксовані та відкриті рішення

33.1. Уже зафіксовано

- Product working name: Lingvessa.
- English і Čeština are launch target languages; Ukrainian is base UI/explanation locale.
- British English default; American English selectable and shown only where meaningful.
- Own code/design/content; no Babbel copying.
- Required media is system-produced/pre-generated before publish; learner does not source it.
- No PDF import in v1.
- Responsive Web/PWA first; content-driven reusable engines; PostgreSQL relational knowledge edges; modular monolith + workers.
- Two-level mastery: six global skills + knowledge-item facets.
- Pronunciation: binary learner-facing feedback only for usable speech; speech_unavailable neutral/non-blocking.
- AI for Learning + Administration; Manual/AI Assist/AI Auto; AI Auto allowlist; core without AI.
- Paid AI budget = 0 by default; free/local-first, provider-neutral.
- Architecture supports professional tracks; full PPC content may follow core stabilization.
- Source code in private Git repository; GitHub Private is current development default.

33.2. Потребує окремого рішення до named stage/public launch

Рішення	Рекомендований default	Коли зафіксувати
Brand/trademark clearance	Use Lingvessa as working name; legal/market clearance before public branding.	Before public/high-fidelity brand launch
Hosting region/provider	Portable managed services; choose by geography/privacy/budget.	Before production environment
Exact image/TTS/STT/AI providers	Provider-neutral adapters; free/local-first candidates evaluated by terms/privacy/quality.	Before relevant integration
Raw voice retention	Do not retain by default.	Before speech beta if exception desired
Launch countries + age flow	Legal/privacy review.	Before real-user registration
Payments/subscription	Separate commercial specification.	Before paid launch
AI paid budget	Remain 0 until owner explicitly approves paid inference.	Before any paid provider use
AI conversation retention	Minimal/off by default; define before public AI conversation.	Before learner AI public use
Native apps/full offline	Defer until web/PWA metrics show need.	Post web stabilization

34. Підсумкове рішення v1.7

Система, яку потрібно будувати
Lingvessa - одна multilingual content-driven learning platform з English і Čeština як двома стартовими мовами, українським UI, окремими curriculum/progress/mastery/review profiles та спільними reusable engines. Mastery має два рівні: six global skills + item facets. Required images/audio генеруються/готуються й QA-перевіряються до publish. Pronunciation gives simple correct/try again only when speech result is usable; unavailable/low-confidence is neutral. AI is an optional provider-agnostic capability layer for Learning + Administration, free/local-first with paid budget 0, and cannot become hidden dependency or bypass governance. Production baseline remains portable: private Git source, Web/PWA, modular API, PostgreSQL, object storage/CDN, workers and adapters.

Додаток A. Зведений перелік core activity types

Група	Types	Призначення
Intro	vocabulary_intro, sentence_intro, scene_intro	Meaning/form/context with audio/media
Selection	multiple_choice, image_choice, classification	Recognition/discrimination
Mapping	matching, pair_match	Form-meaning, audio-text, category
Construction	gap_fill, word_builder, sentence_order	Controlled reconstruction
Production	full_typing, translation, writing	Independent recall/constructed response
Receptive	listening_gist, listening_detail, reading	Comprehension
Dialogue	dialogue_completion, dialogue_chain	Turn-taking/context choices
Speech	listen_repeat, read_aloud, role_play	Perception/articulation/interaction
Grammar	interactive_grammar, contrast, error_correction	Noticing/rule building/form-meaning-use
Assessment	integrated_task, module_recap, level_test	Transfer/coverage/evidence
AI optional	ai_explain, ai_hint, ai_practice, ai_writing_feedback, ai_conversation	Optional grounded support; never core dependency

Додаток B. AI permission matrix

Action	Manual	AI Assist	AI Auto
Publish / rollback	Human admin	AI may prepare preflight summary	Prohibited
Delete/disable account	Human authorized	AI may explain scope	Prohibited
Change roles/MFA/security	Human authorized	AI may recommend	Prohibited
Change AI provider/budget	Human authorized	AI may recommend	Prohibited
Draft lesson/exercise	Human	AI draft	Allowed only in draft scope if allowlisted
Low-risk metadata fix	Human	AI proposal	Allowed with diff/audit

Action	Manual	AI Assist	AI Auto
Create issue from anomaly	Human	AI proposal	Allowed
Generate media prompt/script	Human	AI proposal/job prep	Allowed subject to media gates

Додаток C. Нормативні та внутрішні джерела

Тип	Джерело	Роль / посилання
Internal	Technical Specification v1.7	Implementation blueprint; synchronized with the current candidate baseline.
Internal	Curriculum Blueprint A1 v1.6	A1 curriculum; synchronized with the current mastery/pronunciation contracts.
Internal	Reference Module Design v1.6	Bilingual reference modules; synchronized with current speech/mastery/result UX contracts.
Internal	Technical Design v1.7	Governing corrected implementation contract for two-level mastery and binary speech UX.
Internal	AI Integration Specification v1.4	Governing AI capability layer, free/local-first, budget and permissions.
External	Council of Europe CEFR Companion Volume (2020)	source
External	W3C WCAG 2.2	source
External	OWASP ASVS	source
External	GDPR	source
External	Next.js PWA guide	source
External	PostgreSQL documentation	source

Додаток D. Контрольний чекліст перед програмуванням

- Owner confirms Formal Specification v1.7 and AI Integration Specification v1.4 as the corrected product baseline after audit remediation.
- Technical Specification, Curriculum Blueprint and Reference Module Design are updated to remove legacy mastery/speech wording and point to Formal v1.7.
- Private Git repository is created; repository contains no secrets; CI bootstrap is reproducible.
- Canonical language-neutral content/activity schemas and representative English + Czech fixtures are frozen for first implementation slice.
- Two-level mastery schema (global skill projections + item facets) is explicit in DB/API/tests.
- Speech state machine implements recording pulse, usable correct/try-again, and neutral speech_unavailable fallback.
- AI Layer is disabled-safe, provider-neutral, paid budget 0, and permissions prevent publish/delete/role/billing by AI.
- Media generation manifest proves required images/audio can be produced/stored by tooling without learner sourcing.
- Knowledge taxonomy/edge types, versioning, publish/rollback and immutable attempts are ready before bulk content generation.
- Accessibility, privacy, threat model, originality and QA plans have owners.
- Programming starts with production-oriented bilingual reference implementation, not a disposable one-folder prototype.

КІНЕЦЬ СПЕЦИФІКАЦІЇ v1.7

Цей документ замінює Formal Specification v1.6 як безпосередню попередню продуктову базову версію. Попередні PDF зберігаються для історії змін; наступні Technical/Curriculum/Reference документи мають бути синхронізовані з цією версією та AI Integration Specification v1.4.

Controlled baseline synchronization - 09.09.2026

Normative baseline
This document is part of the synchronized Lingvessa owner-locked implementation baseline; P0 is authorized as of 10.09.2026. Earlier current-version references are superseded by the exact set below; historical "replaces" statements remain historical only.

Document	Version	Authority
Lingvessa_Formal_Specification	v1.7	Product scope, pedagogy, user-facing

		invariants
Lingvessa_Technical_Specification	v1.7	Normative technical requirements and interfaces
Lingvessa_Technical_Design	v1.7	Concrete implementation contracts, DDL/state/API design
Lingvessa_Curriculum_Blueprint_A1	v1.6	A1 pedagogical scope and module architecture
Lingvessa_Reference_Module_Design	v1.6	Executable E-A1-M01/C-A1-M01 fixtures
Lingvessa_AI_Integration_Specification	v1.4	AI-specific policy, data, routing, permissions
Lingvessa_Implementation_Plan	v1.4	Sole execution-order authority; no scope override

Decision	Topic	Fixed rule
BL-001	Baseline set	Formal 1.7 / Technical 1.7 / Design 1.7 / Curriculum A1 1.6 / Reference 1.6 / AI 1.4 / Plan 1.4
BL-002	Precedence	Domain-specific governing document controls; unresolved conflict -> owner decision + ADR/spec revision before code merge.
BL-003	Identity contract	id = UUID; stable_code = immutable human-readable semantic code; version_code = immutable revision label. Never put stable_code in *_id.
BL-004	Mastery contract	Exactly six global skills; canonical facet registry in this release. Reading is assessment-only domain, not seventh mastery skill.
BL-005	Language isolation	Composite DB FKs/checks/trigger guards enforce user + language_profile + language lineage for learner state and learner AI.
BL-006	Pronunciation states	speech_correct / speech_incorrect / speech_unavailable are canonical. Incorrect is non-blocking; Auto-advance occurs only after feedback dwell and never records success.
BL-007	Speech lifecycle	Asynchronous speech_job lifecycle with idempotent final attempt settlement and transient audio cleanup.
BL-008	AI API/data model	Canonical /api/v1/ai/explain|hint|practice|writing-feedback|conversations plus unified ai_* physical model.
BL-009	Monorepo	One /packages/provider-adapters package with ai/, speech/, tts/, image/, email/, storage/ sub-adapters; /api-client and /ui retained.
BL-010	Execution order	Implementation Plan v1.4 P0-P13 is the sole milestone numbering authority.
BL-011	Learner AI opt-in	Learner AI is OFF by default until explicit opt-in/consent. Effective enablement order is global -> capability -> provider/policy -> learner setting/consent -> quota.
BL-012	OpenAPI	Pin API description to OpenAPI Specification 3.1.2 for v1 baseline; upgrades require ADR/contract diff.

Precedence: Formal controls product/pedagogical scope; Curriculum controls A1 pedagogy; Technical Specification controls normative technical requirements; Technical Design controls concrete implementation detail; AI Specification controls AI-specific behavior; Reference Module Design is the executable fixture; Implementation Plan controls execution order only. Any unresolved conflict blocks implementation until an owner decision and controlled revision/ADR.

Canonical mastery/facet registry

Machine-code rule
Only the codes in this registry may appear in mastery_facet/evidence_map. Human labels such as “gist”, “detail”, “target feature”, “repair”, or directional recall must map to these exact machine codes.

facet code	Meaning
recognition	Recognition of a known form/choice
meaning	Meaning comprehension
recall_l2_to_l1	Target-language -> explanation-language recall
recall_l1_to_l2	Explanation-language -> target-language recall
production	Independent target-language production

spelling	Spelling of lexical/form surface
orthography	Language-specific orthographic correctness
listening_recognition	Recognition from audio
listening_gist	Gist from audio
listening_detail	Specific detail from audio
pronunciation_target	Named pronunciation feature
intelligibility	Understandability of learner speech
context_use	Appropriate use in context
collocation	Chunk/collocation integrity
grammar_form	Grammar form
word_order	Word order
morphology	Inflection/morphology
register	Register/formality
variant_awareness	en-GB/en-US or accepted language variant awareness
interaction_repair	Repair/clarification move
interaction_goal	Goal completion in interaction

Facet-to-global-skill aggregation

Global skill	Eligible facets (only when evidence_map explicitly targets the skill)
Vocabulary	recognition, meaning, recall_l2_to_l1, recall_l1_to_l2, production, spelling, context_use, collocation, variant_awareness, register
Grammar	grammar_form, morphology, word_order, register, context_use
Listening	listening_recognition, listening_gist, listening_detail, meaning
Speaking	production, context_use, register, interaction_repair, interaction_goal, intelligibility
Writing	production, spelling, orthography, grammar_form, context_use, register
Pronunciation	pronunciation_target, intelligibility

Reading is not a seventh global mastery skill. Reading tasks may emit item-facet evidence to Vocabulary and/or Grammar according to the objective. The optional Level Test may report a separate assessment_domain=reading subscore; that subscore is assessment-only and is not persisted as skill_mastery_states.

Canonical identity and version contract

Field	Type	Example	Rule
id	UUID	018f...	Opaque database/API identity.
stable_code	string	E-A1-M01-L01	Immutable human-readable semantic code; never stored in *_id.
version_code	string	1.4 / 2026-09-05.1	Immutable revision label for a versioned entity.
provider_external_id	string nullable	vendor-model-x	External/provider identifier only in adapter/config boundary.

API/JSON examples MUST use UUID-shaped values in *_id fields. Content manifests MAY use stable_code for authoring references, but the build/publish step resolves them to UUID + version_code and rejects unresolved codes.

Canonical pronunciation state and asynchronous speech-job contract

State	Meaning	Completion / evidence / Auto-advance
speech_correct	Usable, sufficiently confident target-feature match	Positive bounded Pronunciation evidence. Auto-advance allowed after feedback dwell if setting ON; otherwise show “Далі”.
speech_incorrect	Usable, sufficiently confident mismatch	Bounded negative/practice-needed evidence only for targeted pronunciation facet. Non-blocking. Auto-advance may proceed after feedback dwell if setting ON; otherwise show Retry + Continue. Never counts as success.
speech_unavailable	Mic denied/absent, timeout, low confidence, provider failure	No mastery update. Non-blocking. Auto-advance may proceed after neutral status dwell if setting ON; otherwise Retry + Continue.

Canonical state names are speech_correct, speech_incorrect and speech_unavailable. Legacy labels speech_feedback_correct and speech_feedback_retry are invalid.

speech_job lifecycle

created -> uploaded -> queued -> processing -> succeeded | unavailable | failed | cancelled | expired. Network retries reuse the same attempt_id/idempotency key and existing job; internal provider retries create no learner attempt. succeeded settles exactly once as speech_correct/speech_incorrect; unavailable settles exactly once as speech_unavailable; failed/expired settle as speech_unavailable after retry policy is exhausted. cancelled settles the associated attempt exactly once during /cancel as speech_unavailable(reason=user_cancelled); Continue only advances LessonPlayer after that terminal outcome and creates no new attempt or evidence. Learner Retry after any terminal learner-facing result creates a NEW attempt_id and speech_job. Late/duplicate callbacks are signature-validated, checked against final settlement, logged and ignored after settlement. Client polls GET /api/v1/speech/jobs/{job_id}; SSE MAY be added later. Temporary raw audio is encrypted/transient and deleted immediately after terminal settlement or TTL (maximum 15 minutes by default).

Canonical AI API and persistence mapping

API	Purpose
POST /api/v1/ai/explain	Grounded explanation
POST /api/v1/ai/hint	Bounded hint
POST /api/v1/ai/practice	Extra practice
POST /api/v1/ai/writing-feedback	Writing feedback
POST /api/v1/ai/conversations	Create bounded conversation session
POST /api/v1/ai/conversations/{session_id}/messages	Conversation turn
DELETE /api/v1/ai/conversations/{session_id}	Delete/close learner AI session
Logical concept	**Physical entity/table**
AIProviderProfile	ai_provider_profiles
AIModelProfile	ai_model_profiles
AIPolicy / permissions / budget eligibility	ai_policies
AIPromptTemplate	ai_prompt_templates
AIRequestRecord / AIExecution / AIRun	ai_runs
AISession	ai_sessions
AIMessage	ai_messages
AIArtifact	ai_artifacts
AIUsage / budget consumption	ai_usage_buckets + audit_events
AIToolAction	ai_tool_actions

Learner AI is OFF by default. Effective enablement order: global AI flag -> capability flag -> approved provider/model/data policy -> learner opt-in and required consent -> quota/budget. Any failed check returns deterministic fallback; no paid fallback is implicit.

Canonical execution authority and milestone map

Milestone	Authority
P0	Governance freeze, exact baseline, decision log, full MUST trace matrix
P1	Monorepo/bootstrap/CI/config
P2	Data, UUID/stable_code/version_code, language isolation, publication
P3	Lesson Engine + attempts + deterministic state
P4	Two-level Mastery + Review + mistakes + collections
P5	Media/audio/speech + async speech jobs
P6	English E-A1-M01
P7	Czech C-A1-M01 + bilingual isolation proof
P8	Admin/CMS + publication + Admin AI Assist foundation
P9	Optional learner AI + AI Auto allowlist
P10	Onboarding/Home/Explore/Course/Progress/Profile/Settings/ notifications + persistence
P11	Reference hardening + 32 scenarios + technical gates; freeze reference baseline
P12	A1 M02-M12 controlled rollout + pilot/calibration + optional A1 Level Test
P13	Full-system hardening/launch gate: security, privacy, performance, restore/DR, release provenance, legal/product readiness

Only Lingvessa_Implementation_Plan v1.4 uses P0-P13 numbering. Any older P-number labels in other specifications are descriptive history and MUST be mapped to these milestones rather than treated as execution authority.

Canonical external references

- OpenAPI Specification 3.1.2
- CEFR Companion Volume

- Council of Europe - CEFR
- English Profile
- Čeština pro cizince

OpenAPI here means the OpenAPI Specification for describing HTTP APIs; it is unrelated to OpenAI as an AI provider. OpenAI remains only a possible future provider behind the provider-agnostic AI adapter and is not required by the v1 baseline.

Approval record

Поле	Значення
Baseline status	OWNER-LOCKED IMPLEMENTATION BASELINE / P0 AUTHORIZED
Approval date	10.09.2026
Approver	Product owner
Candidate version	v1.7
Accepted exceptions	None
Execution authority	Lingvessa_Implementation_Plan v1.4; P0 must be closed before P1.

Canonical machine-readable source: /packages/activity-schemas/mastery-registry.json. CI MUST validate every evidence_map against the registry and eligible skill-facet pair matrix; copied tables in specifications are checksum-checked mirrors.

Perceptual pronunciation-awareness tasks (for example selecting the stressed syllable) map to Listening/listening_recognition. Pronunciation mastery evidence requires usable learner audio; typed role-play never creates Speaking/Pronunciation evidence, though it MAY also create explicitly mapped Grammar/form/register evidence when the written response assesses that objective.